

Data Visualization

Computational Statistics Team Teaching
2024/2025

Outlines



- One-Way Data
- Visualize One-Way Data
- Two-Way Data
- Visualize Two-Way Data

One-Way Data

Analyzing a single factor or variable and its effect on a response variable.

Example One-Way Data

Dependent Variable (DV) / Individuals

Independent Variable (IV) / Variable

Class	Members
TI-2A	30
TI-2B	28
TI-2C	29
TI-2D	30

What if, if we have more than 1 variabels?



Name	Sex	Age	Class
Amar	L	19	TI-2A
Bagus	L	19	TI-2B
Citra	P	19	TI-2C
Dannise	P	19	TI-2D

- One-way data involves analyzing a **single factor** or variable and its effect on a response variable.
- If you have additional variables used to describe or identify your **data groups**, those are not considered "factors" in the analysis.

Another Example One-Way Data

What if the IV > DV?

	House Type A	House Type B	House Type C
Building Size	36m ²	40m ²	54m ²
Lot Size	60m ²	84m ²	90m ²
Num. Bedroom	2	2	3
Num. Restroom	1	1	2
Parking Lot	2x3 m	2x3 m	2x3 m
Annual Tax	2jt	2.5jt	3jt
Garden	Has	Has	Has
Basement	-	-	Has
Electrical Power	1300	1300	2400

One-Way Data: Consideration

- A one-way analysis can include multiple variables as long as these variables represent repeated measurements or observations under the same factor.

Diet Type	Weight Loss (kg)	Calorie Reduction (%)
Diet A	2.5	15
Diet B	3.1	20
Diet C	4.2	25

DV

One-Way Data: Consideration

- If you have additional variables used to describe or identify your **data groups**, those are not considered "factors" in the analysis

Participant	Diet Type	Gender	Weight Loss (kg)
1	Diet A	Male	2.5
2	Diet B	Female	3.1
3	Diet C	Male	4.2

Categorical -> Grouping data

How to visualize one-way data

FIFA World Cup Hosts: 1970 – 2002 (1)

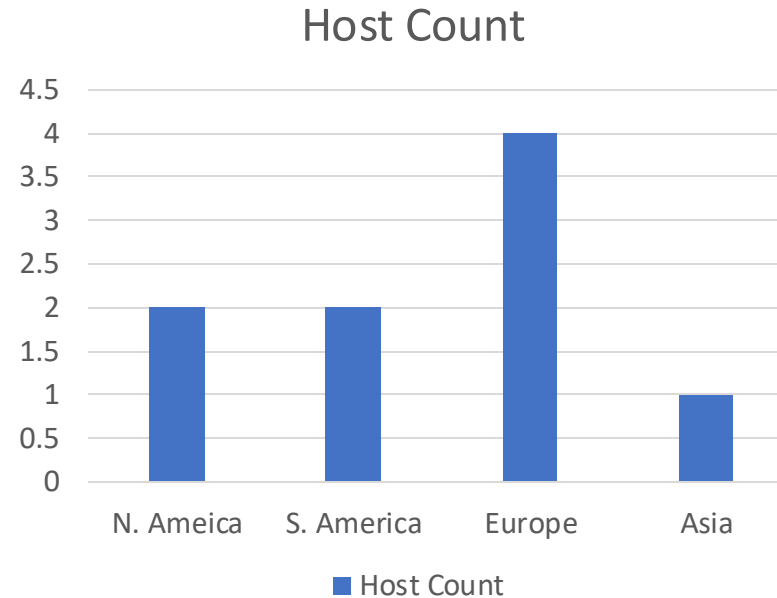
Year	Host Nation	Continent
1970	Mexico	N. America
1974	West Germany	Europe
1978	Argentina	S. America
1982	Spain	Europe
1986	Mexico	S. America
1990	Italy	Europe
1994	United States	N. America
1998	France	Europe
2002	Japan S. Korea	Asia



Continent	Host Count
N. America	2
S. America	2
Europe	4
Asia	1

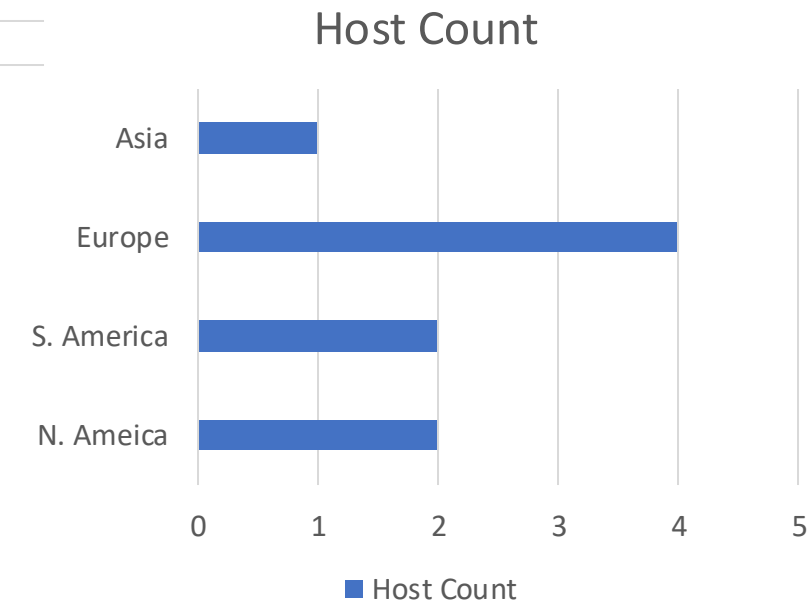
FIFA World Cup Hosts: 1970 – 2002 (2)

Continent	Host Count
N. America	2
S. America	2
Europe	4
Asia	1



**Vertical
Bar Chart**

**Horizontal
Bar Chart**

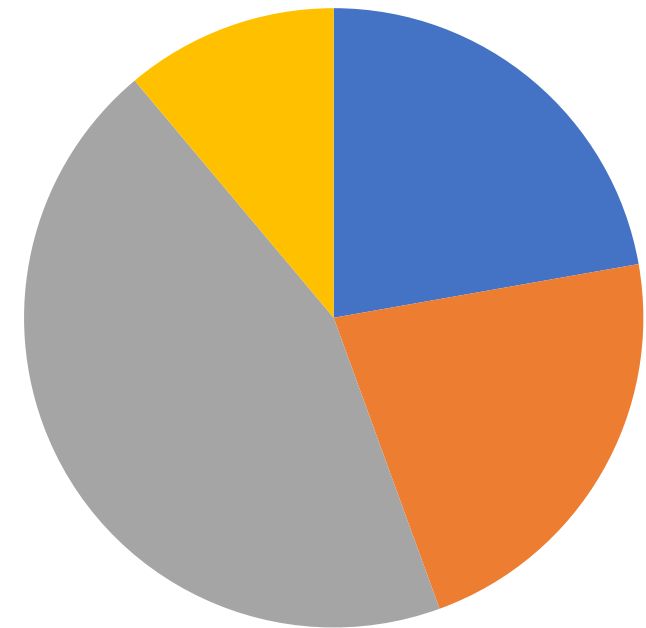


FIFA World Cup Hosts: 1970 – 2002 (3)

Pie Chart

Host Count

Continent	Host Count	%
N. America	2	~22%
S. America	2	~22%
Europe	4	~44%
Asia	1	~12%



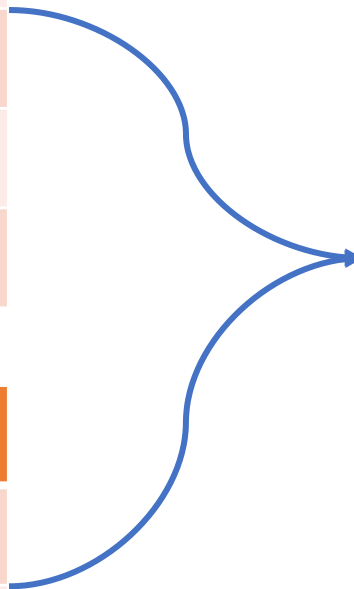
■ N. Ameica ■ S. America ■ Europe ■ Asia

Other Example: Summer and Winter Olympics Hosts (1)

Continent	Host Count
Europe	16
N. America	6
Asia	3
Australia	2
S. America	1

Continent	Host Count
Europe	14
N. America	6
Asia	3

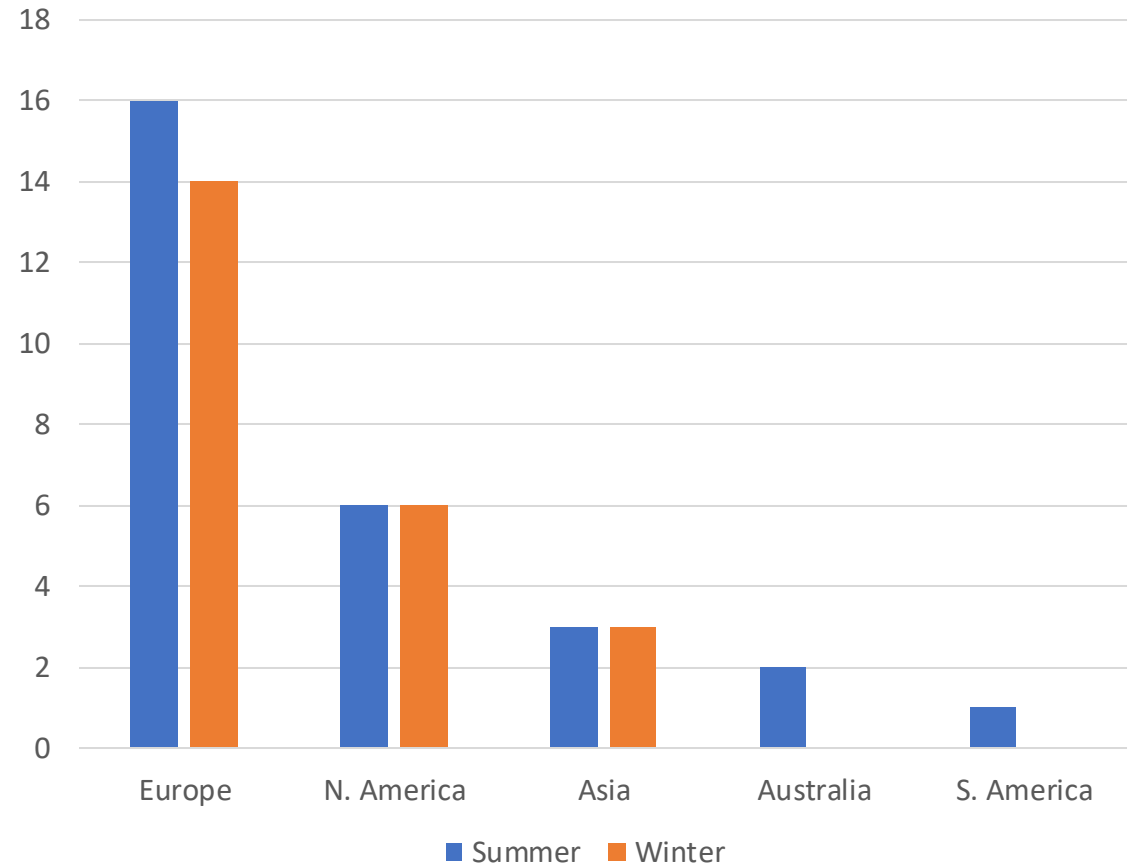
Continent	Summer	Winter
Europe	16	14
N. America	6	6
Asia	3	3
Australia	2	0
S. America	1	0



Other Example : Summer and Winter Olympics Hosts (1)



Continent	Summer	Winter
Europe	16	14
N. America	6	6
Asia	3	3
Australia	2	0
S. America	1	0



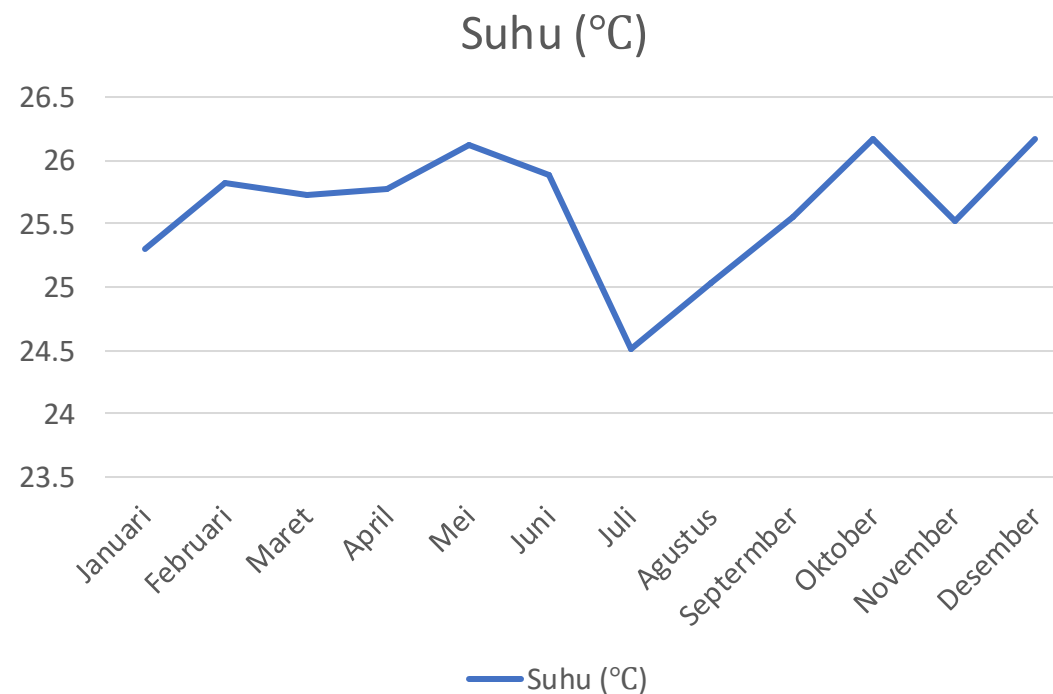
Is it still a one-way data?

Another Example: Temperature in Malang 2021

What kind of chart
which suitable?



TREND!



Line Chart

Best Practice



If it discrete / Showing Amount
/ Ratio → Bar Chart / Pie Chart

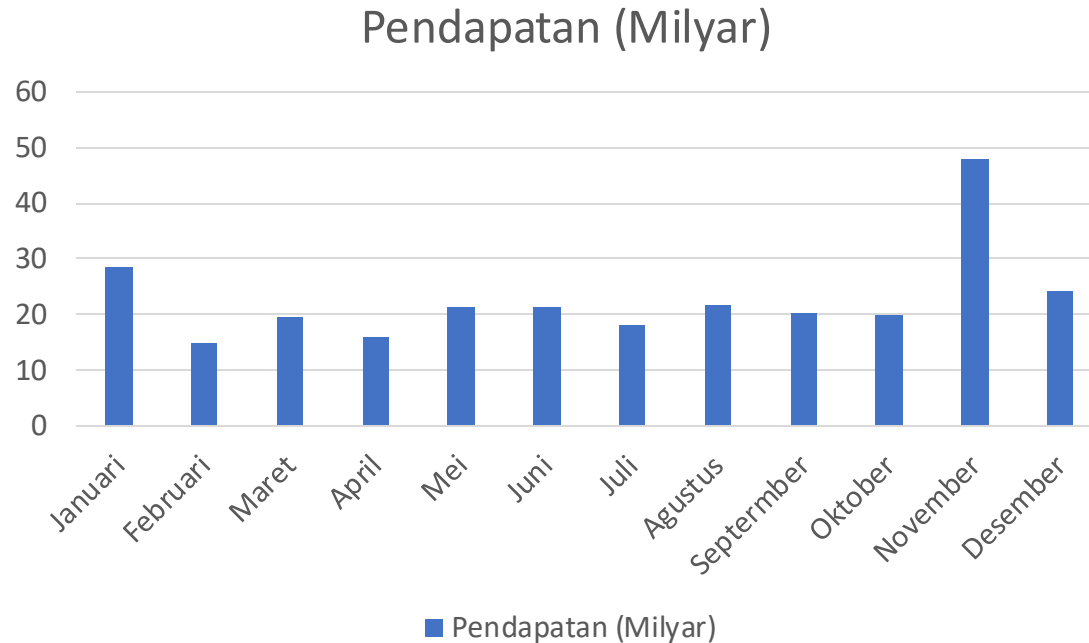


If it showing a trend → Line
Chart

Two-Way Data

Analyzing two factors (or independent variables) and their effects on a response variable. It also considers how the two factors interact with each other.

Bulan	Pendapatan (Milyar)
Januari	28.361
Februari	14.744
Maret	19.407
April	15.891
Mei	21.277
Juni	21.530
Juli	17.990
Agustus	21.838
September	20.174
Oktober	20.025
November	48.055
Desember	24.318



- ❖ Only show specific information
- ❖ Doesn't show the trend of the specific month



Another Example: PT. XYZ Revenue

Month	2014	2015	2016	2017	Total
JAN	1.396	10.696	16.015	28.361	
FEB	1.084	4.745	8.538	14.744	
MAR	7.549	9.888	19.407		
APR	2.434	10.643	15.891		
MEI	0.820	2.708	2.326	15.423	21.277
JUN	1.138	3.014	2.982	14.398	21.530
JUL	0.694	2.586	3.270	11.440	17.990
AGU	0.486	2.080	5.451	13.821	21.838
SEP	0.538	3.109	6.614	9.913	20.174
OKT	0.448	3.695	5.153	10.729	20.025
NOV	2.387	6.495	13.128	26.045	48.055
DES	0.401	2.030	7.411	14.466	24.318
Total	8.890	31.544	71.789	161.297	273.610



Another Example: PT. XYZ Revenue

Two-way Data: Analyze More Than 1 Factors!

This is the most important things!

- Two-way data involves analyzing two factors (or independent variables) and their effects on a response variable. It also considers how the two factors interact with each other.
- **Example:** If you want to study the effect of **diet type** **AND** **exercise level** on weight loss, where both factors (diet and exercise) are being analyzed, it's two-way data.

Diet Type	Exercise Level	Weight Loss (kg)
Diet A	Low	2.5
Diet A	High	3.8
Diet B	Low	3.1
Diet B	High	4.5
Diet C	Low	4.2
Diet C	High	5.7

Factors

individuals

Visualize Two-Way Data

How to visualize two-way data?

Bar Chart



Revenue PT. XYZ Jaya Sentosa
2014 - 2017

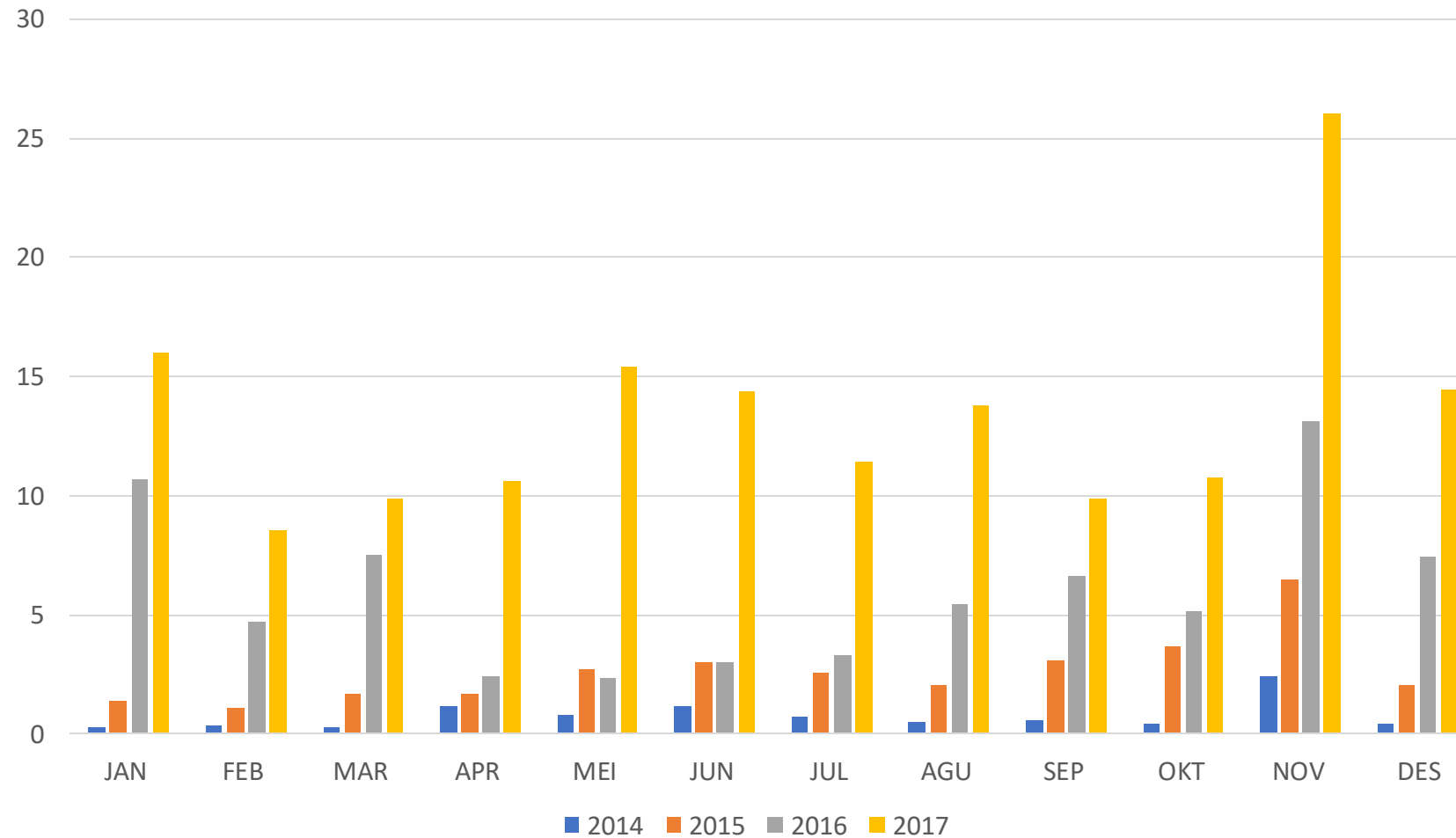
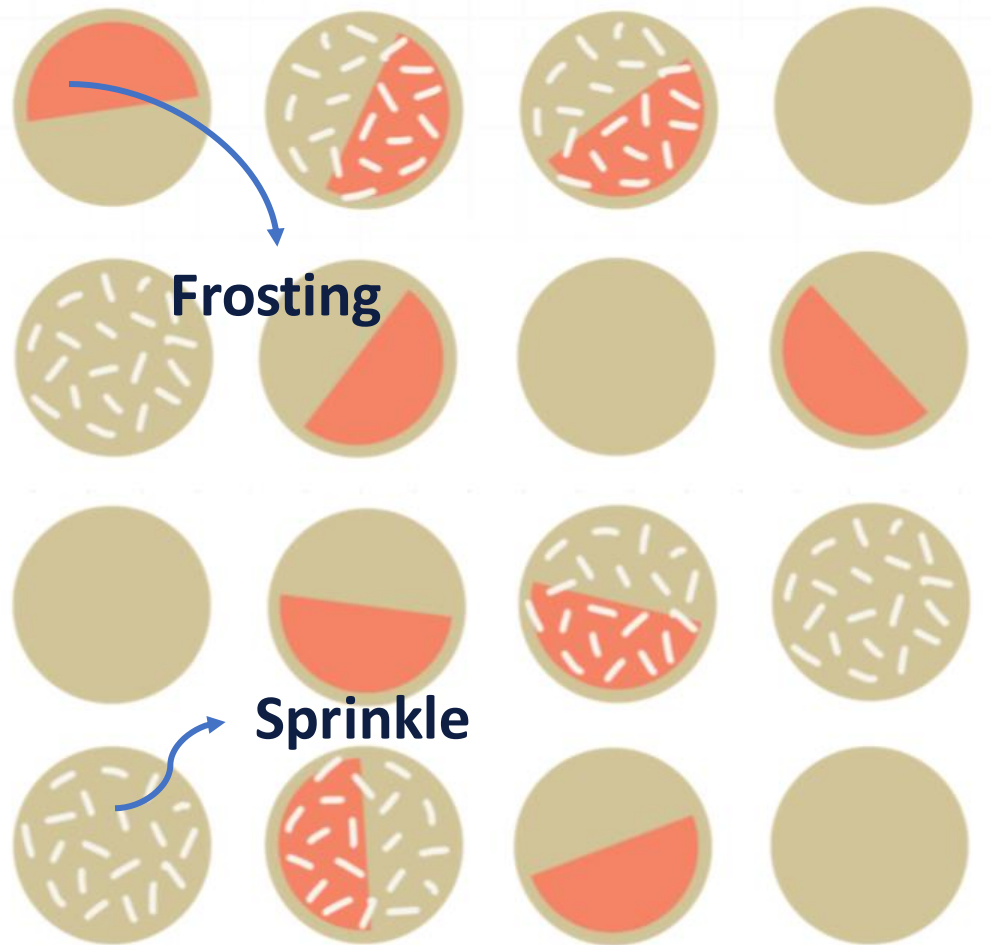
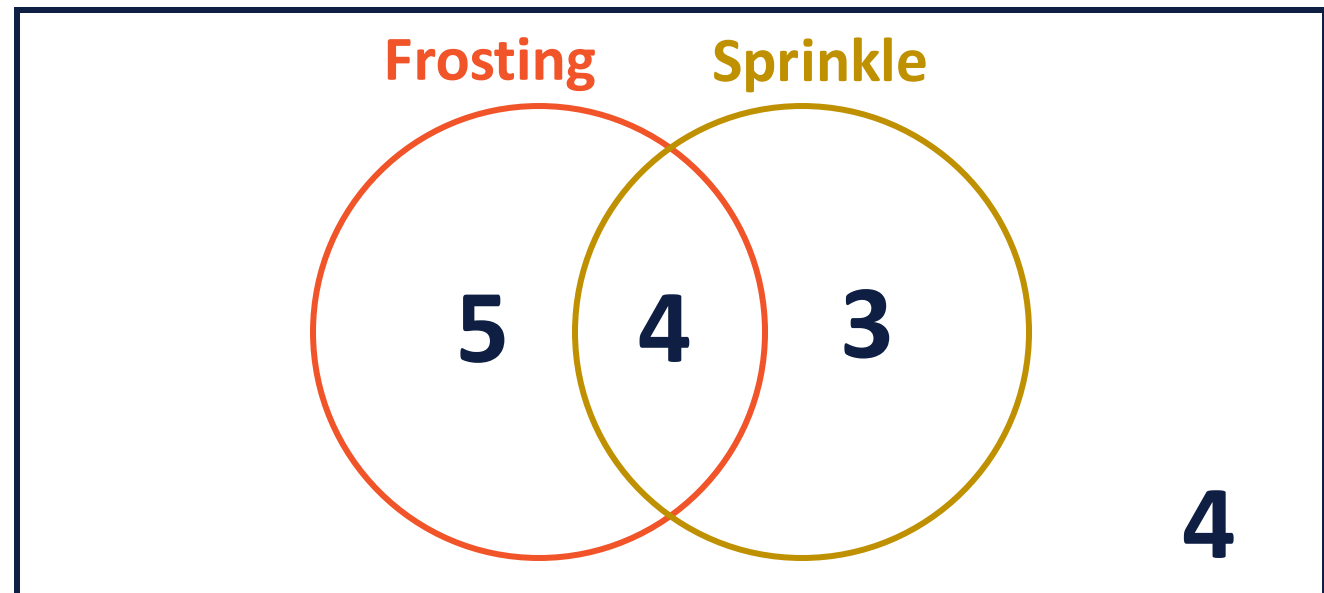


Diagram Venn: *Let's get some cookies!*



	Sprinkle	No Sprinkle	Total
Frosting	4	5	9
No Frosting	3	4	7
Total	7	9	16

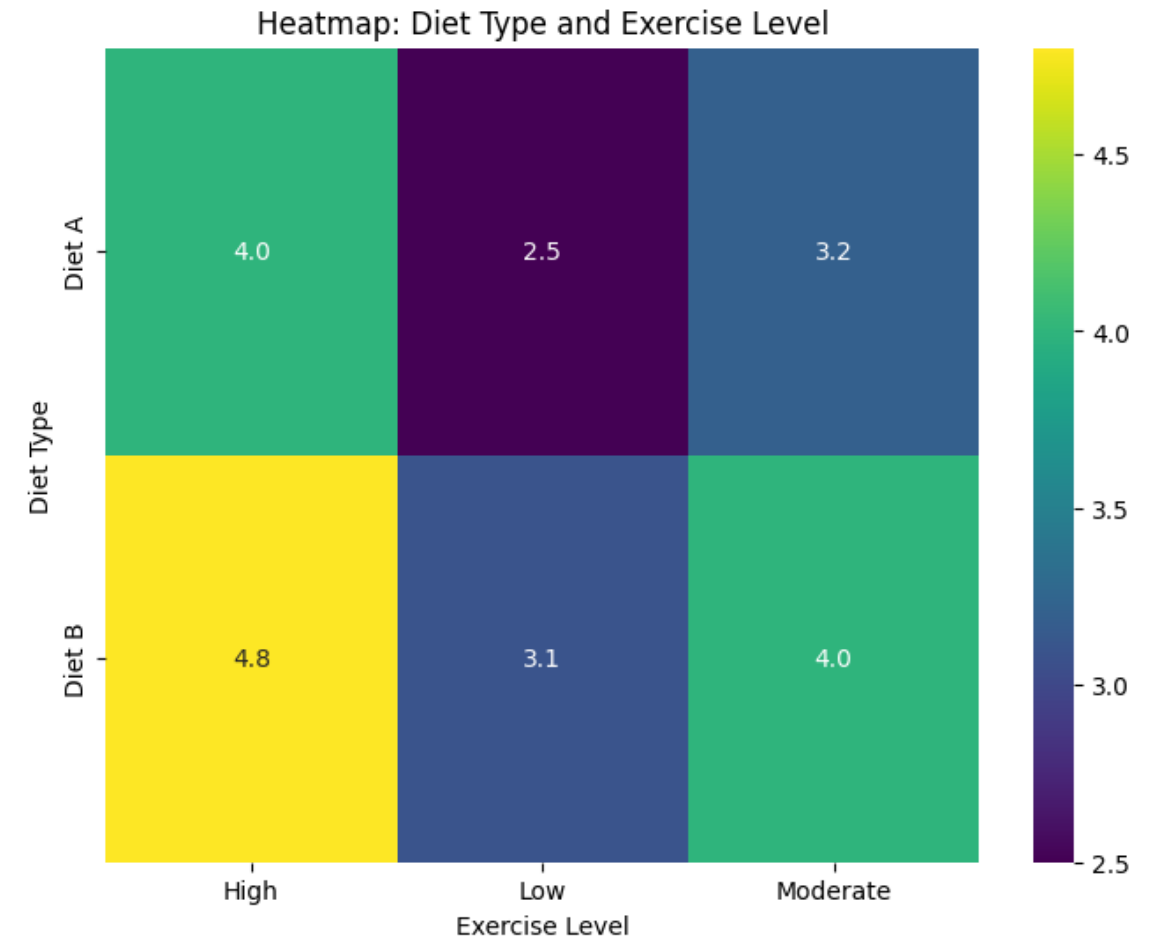
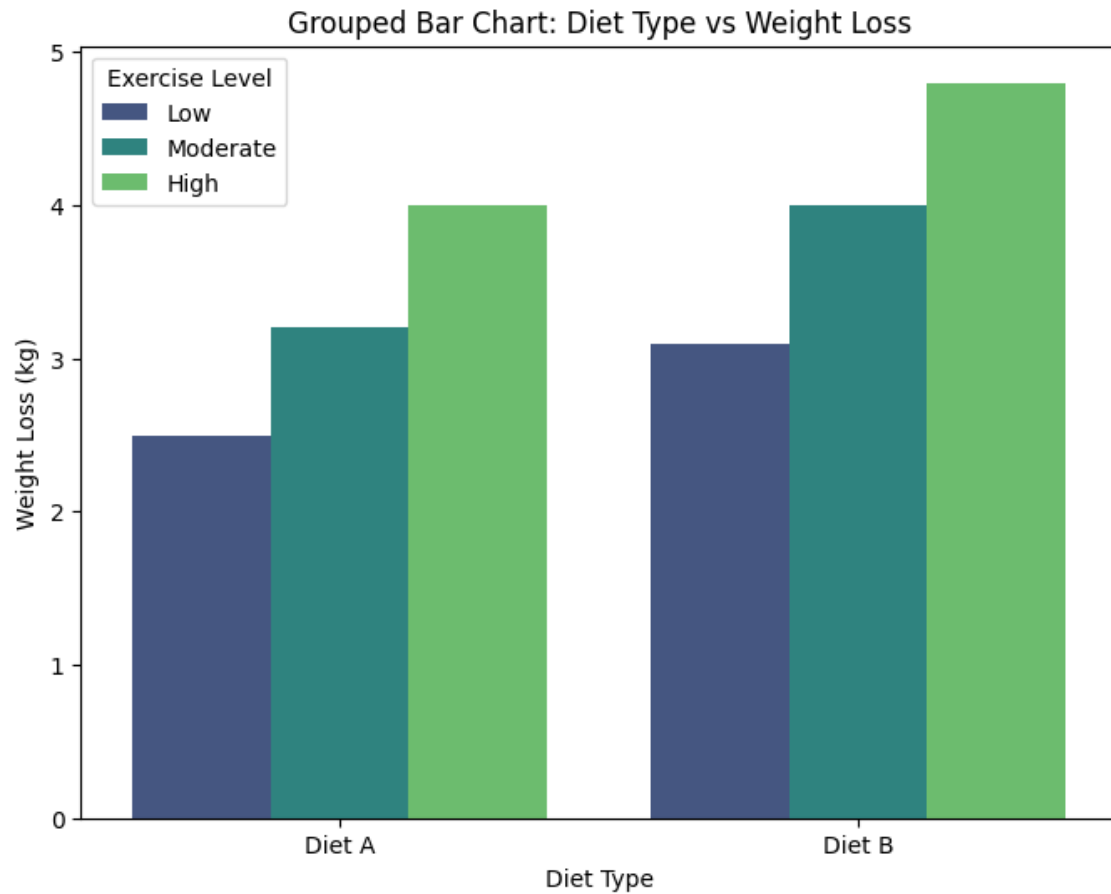


Other Case Example (1)

- A grouped bar chart is ideal for visualizing **the interaction between two factors** when comparing categories of a dependent variable.
- **Example:**
 - **Factors:** Diet Type (A, B, C) and Exercise Level (Low, Moderate, High)
 - **Dependent Variable:** Weight Loss (kg)
 - Each diet type will have grouped bars for different exercise levels.

Diet Type	Exercise Level	Weight Loss (kg)
Diet A	Low	2.5
Diet A	Moderate	3.2
Diet A	High	4.0
Diet B	Low	3.1
Diet B	Moderate	4.0
Diet B	High	4.8

Other Case Example (1)



Another data visualization

Frequency table and histogram

Frequency Table (1)

Survey of Weight of 30 Asian

20	30	35	40	45	50	65	60	70	80
90	85	75	25	33	36	50	80	65	10
95	75	65	55	75	80	77	78	50	48

Step 1

Divide the range (the difference between the largest and smallest observations) of the measurements by the desired number of intervals, which is typically between 5 and 20.

Step 2

Round the number obtained in Step 1 to a convenient unit. The unit represents the general width of the class intervals.

Step 3

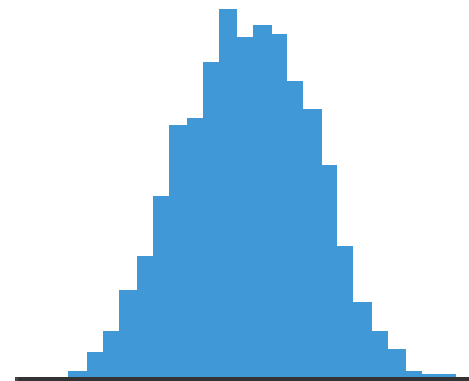
Choose the first class interval such that it includes the smallest measurement. Ensure there are no ambiguous values between two intervals.

Frequency Table (2)

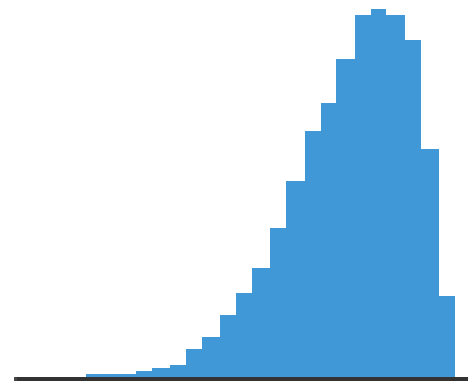
Class Interval	Frequency	Relative Freq.	Cumulative Freq.
10 – 27	3	0.10	0.10
28 – 45	6	0.20	0.30
46 – 63	6	0.20	0.50
64 – 81	12	0.40	0.90
82 – 99	3	0.10	1.00
Total	30	1.00	

Caution!
There is no standard
to decide the interval

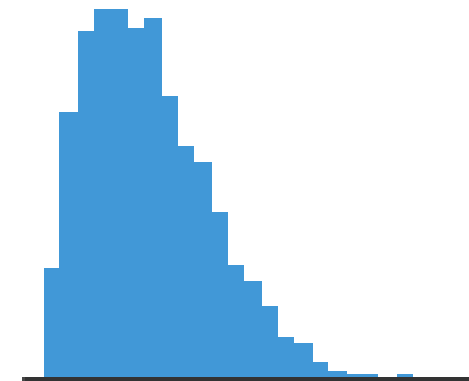
Histogram



symmetric, unimodal



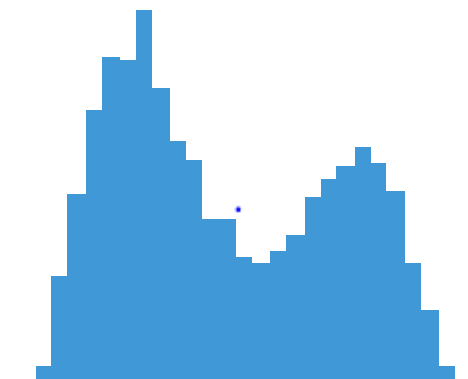
skew left



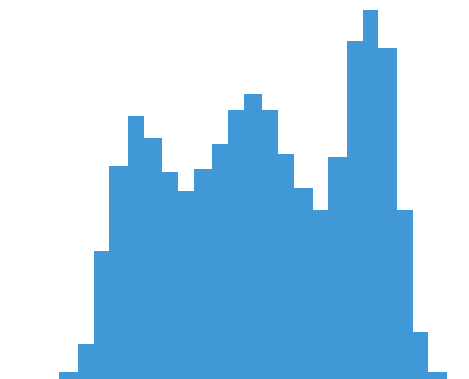
skew right



uniform



bimodal



multimodal

Sumber: <https://chartio.com/learn/charts/histogram-complete-guide/>

How to Make a Histogram

- Use intervals of equal length.
- Show the entire vertical axis starting from 0.
- Avoid breaking or splitting the axis.
- Place bars with appropriate heights at the midpoint of each interval.

Example: Histogram

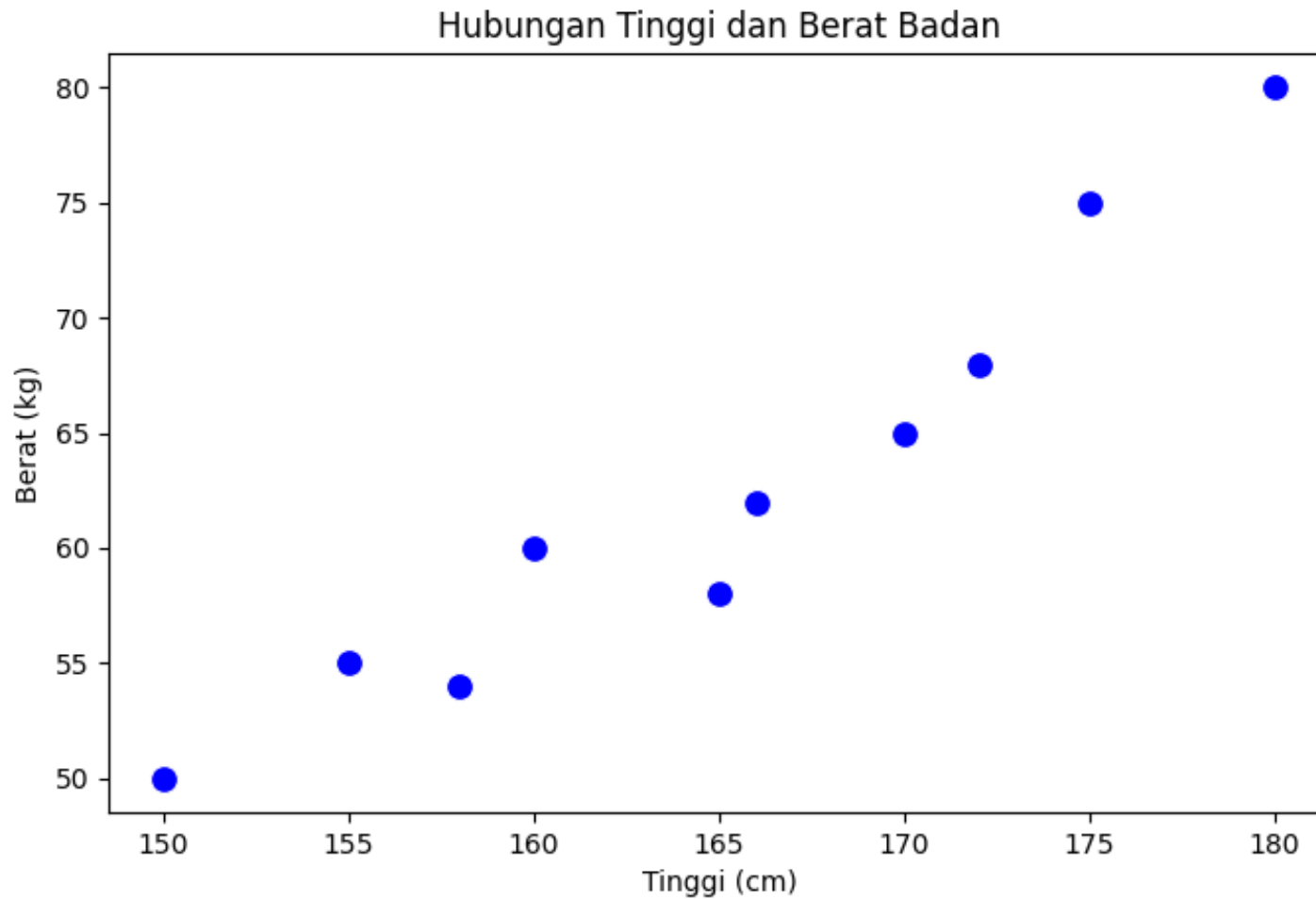
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Total	30	1.00	



Data Visualization Using Python

<https://colab.research.google.com/drive/1whXOpVIPeRzN9C-wSfxKyjJhWvU1BWCR?usp=sharing>

Extra: Scatter Plot



Practice!

- Create a freq. table and histogram based on this following data

21	20	31	24	15	21	24	18	33	8
26	17	27	29	24	14	29	41	15	11
13	28	22	16	12	15	11	16	18	17
29	16	24	21	19	7	16	12	45	24
21	12	10	13	20	35	32	22	12	10

